

MD. ZAKIR HOSSAN

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Career Goal

To become a pioneering academic figure, driving cutting-edge research, inspiring innovation, and fostering the next generation of tech leaders, while creating a dynamic learning environment that fuels unprecedented advancements.

Research Interests

- Machine Learning
- Physics-Informed Neural Nets
- Computer Vision
- Quantum Computing
- Blockchain
- Control Theory

Ongoing Research

Advancing High-Energy Particle Physics through Machine Learning for Enhanced Discovery and Analysis, under the supervision of Professor **M Ashraful Amin** and Professor **Arshad Momen**.

Physics-Informed Machine Learning, under the supervision of Professor **Arsad Momen** and Professor **Amin Ahsan Ali**.

Experience

Research Experience

Research Manager

Center for Computational & Data Sciences, CCDS

Jan 2022 – until the present

Independent University, Bangladesh

Research Associate

Artificial Intelligence and Cybernetics Lab

Jan 2020 – Dec 2021

Independent University, Bangladesh

Research Assistant

Computer Vision & Cybernetics Group

Jan 2016 – Dec 2019

Independent University, Bangladesh

Teaching Experience

Lecturer

Sonargaon University

May 2018 – Apr 2022

Adjunct

Lecturer

Manarat International University

Jun 2021 – Feb 2022

Adjunct

Instructor

National Institute of Engineering & Technology

Feb 2014 – Apr 2017

Lecturer

MCET Engineering College

May 2012 – Sept 2013

Industry Experience

Software Engineer

ISHO Ltd

Mar 2019 – Sept 2019

DekkoISHO Group

Deep Learning Engineer

Socian Ltd.

Sept 2017 – Feb 2018

Software Engineer

Business Accelerate BD Ltd.

May 2017 – Aug 2017

R&D

Publications

Journal Paper

- Md Raqibul Islam, Adrita Khan, Mir Sazzat Hossain, Choudhury Ben Yamin Siddiqui, **Md. Zakir Hossan**, Tanjib Khan, M. Arshad Momen, Amin Ahsan Ali, AKM Mahbubur Rahman, "E-PCN: Jet Tagging with Explainable Particle Chebyshev Networks Using Kinematic Features", <https://arxiv.org/abs/2512.07420>.

Conference Paper

- Mohammad Aftab Mahmud, Md. Ridwan Mahmud, **Md. Zakir Hossan**, Asif Mahmud, Ashraful Islam, M Ashraful Amin, "Blockchain-Enabled Voluntary Carbon Credit Marketplace for Bangladesh: A Decentralized Solution to Climate Accountability", 2025 IEEE 23rd Student Conference on Research and Development (SCoReD), 25 - 26 November 2025 - Universiti Malaya, Kuala Lumpur, Malaysia.
- Sayed Islam, **Md. Zakir Hossan**, "A Review of Available Standard Dataset for Automated Medicine Pill Recognition Technology", Third International Conference on Intelligent Computing, Instrumentation & Control Technologies, ICICICT-2022. Aug 11-12, 2022, Kannur, Kerala, India.
- Sayed Islam, Md. Ferdous Khan, **Md. Zakir Hossan**, M Ashraful Amin, "An Overview of Radio over Fiber (RoF) Technology", 2nd INTERNATIONAL CONFERENCE ON INTELLIGENT COMPUTING, INSTRUMENTATION AND CONTROL TECHNOLOGIES. Jul 5-6, 2019, Kannur, Kerala, India.
- Tanjina Piash Proma, **Md. Zakir Hossan**, M. Ashraful Amin, "Medicine Recognition Using Number of Color and Text of Pill Image", 3rd International Conference on Graphics and Signal Processing. June 1-3, 2019, Hong Kong.
- Md. Zakir Hossan**, Sayeed Islam, Md. Ferdous Khan, Sayma Shammi, M. Ashraful Amin, "Boiler Explosion In Bangladesh: Causes, Consequences and Precautions", International Conference on Management Science and Industrial Engineering. May 24-26, 2019, Phuket, Thailand.
- Sakhawat Hosain Sumit, **Md. Zakir Hossan**, Tareq-Al-Muntasir, Tanvir Hassan Sourov, "Sentiment Analysis with Recurrent Neural Network using Long Short-Term Memory in Bengali Language", International Conference on Bangla Speech and Language Processing (ICBSLP), September 21-22, 2018, SUST, Sylhet, Bangladesh.

Datasets

- Recognizing Medicine from Medicine Image:** Develop and deploy the medicine recognition system on the custom dataset and improve the overall feature detection techniques, which will be extremely useful to identify their prescribed pills out of the packaging.
- National Medicine Image Database Creation:** Collecting raw pill image data maintains the quality of images such as proper lighting, angle, and visual of the engraved or imprinted text on the surface.
- 40k Bangla Newspaper Article:** Custom-made Bangla text corpus from various online newspapers. The Dataset may be used for Text Generation, Document Classification, Topic Modeling, etc.
<https://www.kaggle.com/zshujon/40k-bangla-newspaper-article>.
- 4K Bangla Song Lyrics:** The dataset contains 4K (4068) Bangla song lyrics in Bangla Unicode font. All the lyrics were collected via web scraping. Those who want to use this must do some cleaning. The dataset is useful for character-by-character sequence generation or any other NLP task.
<https://www.kaggle.com/zshujon/4k-bangla-song-lyrics>.
- Bangla digits:** Bangla digits from standard TTF (true type font) file.
<https://www.kaggle.com/zshujon/bangla-digits-from-ttf-files>

Education

North South University	2017
<i>Master of Science in Electrical & Electronic Engineering</i>	<i>3.67/4.0</i>

International Islamic University Chittagong	2011
<i>Bachelor of Science in Electrical & Electronic Engineering</i>	<i>3.514/4.0</i>

University Laboratory College	2006
<i>Science, Dhaka Board</i>	<i>3.7/5.0</i>

AK High School	2004
<i>Science, Dhaka Board</i>	<i>4.69/5.0</i>

Online Certification

6.00.1x: Introduction to Computer Science and Programming Using Python	MITx
https://verify.edx.org/cert/98aa9b561646439bb0aa3335a8d21069	<i>edX</i>

Technical Skills

Languages: Python, Java, C, SQL

Developer Tools: VS Code, Eclipse, Google Cloud Platform, Android Studio

Technologies/Frameworks: Linux, GitHub, WordPress

References

M Ashraful Amin, PhD

- Founder Director, Center for Computational & Data Sciences
- Professor, Computer Science & Engineering
- Independent University, Bangladesh
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M Arshad Momen, PhD

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- Professor, Physical Sciences
- Independent University, Bangladesh
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